

# Junghwan Yim

Ph.D. Student · Georgia Institute of Technology  
dlagudah@gmail.com · GitHub: JacobYim · LinkedIn

## RESEARCH INTERESTS

Agentic digital twins; robotics security; sensing and applied machine learning. Developing methods that connect physical observations, virtual models, and robot behavior, with an emphasis on implementation and evaluation.

## EDUCATION

|                                                                                                                                                              |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Georgia Institute of Technology</b><br>Ph.D. in Mechanical Engineering, in progress                                                                       | Jul 2025 - Present |
| <b>University at Buffalo</b><br>M.S., Physics; B.S., Computational Physics; B.S., Computer Science                                                           | 2016 - 2022        |
| <b>State University of New York at Fredonia</b><br>Bachelor's degree, Physics with Pre-Law<br>Bachelor's degree, Business Administration with Music Industry | 2013 - 2016        |

## PROFESSIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Hyundai Motor Group - Meta Assembly</b><br>AI Solution Architect Intern                                                                                                                                                                                                                                                                                                                                                              | May 2026 - Aug 2026 |
| Architected a Transformer world model on an NVIDIA Isaac Sim factory digital twin for AMR routing and logistics. Built a departmental multi-agent LLM platform on Kubernetes with MCP data access. Developed a Neo4j knowledge ontology and real-time consistency validation, reducing the <b>daily production-issue rate by 80%</b> . Owned data pipelines, deployment, GPU scheduling, and observability for <b>24/7 operations</b> . |                     |
| <b>SK C&amp;C USA, Inc.</b><br><b>Senior Solution Engineer - AI   Robotics Specialist</b>                                                                                                                                                                                                                                                                                                                                               | Sep 2024 - Aug 2025 |
| Designed AI, robotics, and digital twin solutions from customer requirements through PoC and production handoff, generating approximately <b>\$35M in potential sales pipeline</b> . Led PROMAT 2025 and digital marketing initiatives with a <b>\$30K budget</b> , securing <b>300+ leads</b> . Architected AMR/AGV, RFID, and inspection integrations with MES/WMS/ERP systems; delivered customer workshops and onboarding.          |                     |
| <b>Automation Project Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                      | Mar 2023 - Aug 2024 |
| Engineered automation for the Ford-SK On battery factory across software, PLCs, vision, and AMR/AGV control. Built a RAG-based engineering-knowledge platform delivering approximately <b>\$20M in cost savings</b> through reduced rework and faster onboarding. Developed AI inspection, drawing generation, 3D installation verification, and logistics simulations.                                                                 |                     |
| <b>Hyundai AutoEver America</b><br>Software Test Engineer                                                                                                                                                                                                                                                                                                                                                                               | Oct 2022 - Mar 2023 |
| Developed AI-assisted navigation testing for the Genesis lineup using sign and vehicle-state recognition. Contributed to HD maps supporting autonomous-driving and ADAS functions.                                                                                                                                                                                                                                                      |                     |
| <b>Republic of Korea Army - Cyber Operation Command</b><br>Software Engineer / Data Scientist                                                                                                                                                                                                                                                                                                                                           | May 2019 - Dec 2020 |
| Architected and operated a <b>petabyte-scale distributed intelligence platform</b> for data collection, NLP classification, and entity resolution using Hadoop, Spark, Kafka, and ELK. Built React/Django interfaces and Celery/RabbitMQ/Redis pipelines. Received <b>2 Meritorious Service Medals</b> for software development and the K-Startup Contest (2019-2020).                                                                  |                     |

## RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                             |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Georgia Institute of Technology</b><br>Graduate Research Assistant                                                                                                                                                                                                                                                                                                       | Aug 2025 - Present  |
| Built multi-node, multi-GPU training pipelines and multiscale simulation frameworks combining learned interatomic potentials with DFT. Developed synchronized LiDAR/camera/radar testbeds. Co-authored SeeTraceAct with Prof. Zsolt Kira's group, fine-tuning VLA action policies on real Franka Panda hardware.                                                            |                     |
| <b>University at Buffalo - Biometrics Lab</b><br>Graduate Student Researcher                                                                                                                                                                                                                                                                                                | Jun 2022 - Jun 2024 |
| Developed authentication models using phone and watch IMU signals, transfer functions, CycleGAN, signal processing, and machine learning. Built phone and watch applications for data collection, Bluetooth-connected sensor monitoring, and model-based authentication.                                                                                                    |                     |
| <b>University at Buffalo - CAVAS Lab</b><br>Graduate Student Researcher                                                                                                                                                                                                                                                                                                     | Jan 2021 - Jun 2024 |
| Led a traffic-sign object detection team and built a traffic-sign knowledge graph. Trained YOLO and RCNN models on Mapillary traffic-sign data, with the best model achieving <b>70% accuracy</b> . Implemented A2C and DQN agents in synchronous and asynchronous CARLA simulation, and collaborated on a ROS package mapping detected signs and distances into an HD map. |                     |

## University at Buffalo - Condensed Matter Physics Lab

Jun 2016 - Jun 2022

Graduate Student Researcher

Modeled electron scattering and graphene transport using the Caldeira-Leggett model and semiclassical Boltzmann transport equation. Implemented Python and C++ models with high-performance computing and MPI; visualized and evaluated models under varying conditions.

## University at Buffalo - Blockchain Lab

May 2017 - May 2019

Graduate Student Researcher

Participated in developing a Solidity DApp for credit-information exchange in an M&T Bank hackathon case.

## TEACHING

### SUNY Fredonia

Aug 2015 - Jan 2016

Teaching Assistant, Physics Laboratories

Guided experiments and advised students on equipment and procedures across **4 physics lab courses**: University Physics I/II and College Physics I/II.

## JOURNAL AND CONFERENCE PUBLICATIONS

- [1] Young-Dae Shim, **Junghwan Yim**, Nian Liu, Cheong Hoon Kwon, Hae-Jin Choi, Seung-Kyum Choi. Cross-Chemistry Meta-Learning with Frequency-Domain Augmentation for Robust Capacity Retention Prediction in Vanadium Redox Flow Batteries. *IEEE Access*, 2026. <https://doi.org/10.1109/ACCESS.2026.3737729>
- [2] Seokmin Choi, **Junghwan Yim**, Se Jun Kim, Yincheng Jin, Di Wu, Zhanpeng Jin. VibPath: Two-Factor Authentication with Your Hand's Vibration Response to Unlock Your Phone. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023. <https://doi.org/10.1145/3610894>
- [3] Seokmin Choi, **Junghwan Yim**, Yincheng Jin, Yang Gao, Jiyang Li, Zhanpeng Jin. EarPPG: Securing Your Identity with Your Ears. *ACM International Conference on Intelligent User Interfaces (IUI)*, 2023. <https://doi.org/10.1145/3581641.3584070>

## WORKSHOP PAPER

- [4] Jaehyeon Son, Junhyun Kim, Kyle Kam, Jinhoo Kim, Seok Joon Kim, **Junghwan Yim**, Zsolt Kira. SeeTraceAct: Visibility-Aware Latent Planning from Cross-Embodiment Demonstration Videos. *SCALE Workshop @ ICML 2026*. <https://openreview.net/forum?id=xZbOwW72u5>

## PREPRINT

- [5] Mainak Mallick, **Junghwan Yim**, Alankrit Gupta, Seung-Kyum Choi. Rad-R: A Raw-ADC Radar Dataset and Capture-Invariant SSM for Hardware-Fault Diagnosis. *arXiv preprint*, 2026. <https://arxiv.org/abs/2608.30896>

## SELECTED ENGINEERING PROJECTS

### Music NFTs and audience engagement

Feb 2022 · Chartmetric · Music analytics · Measure of Music 2022

A Measure of Music conference project examining music NFT releases using Chartmetric data. The analysis considered Shawn Mendes and Tory Lanez releases alongside follower and streaming trends.

### Reinforcement learning agents in CARLA

Mar - May 2021 · CARLA · DQN · A2C

Autonomous driving agents using DQN and A2C in the CARLA simulator, with synchronous and asynchronous simulation environments.

### Rumour detection with BERT and GCN

Feb - May 2021 · BERT · Graph convolutional networks · RumourEval 2019

A model for classifying discussion stances and detecting rumours in Twitter and Reddit data from RumourEval 2019. The project combines BERT with a three-layer graph convolutional network.

Achieved **67% stance-classification accuracy** and **70% rumour-detection accuracy** on RumourEval 2019. Selected as the best study in the Advanced Natural Language Processing class.

SELECTED HONORS AND AWARDS

|                                                                                                                         |                            |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Employee of the Year</b> , SK C&C USA<br>Recognition for AI-aided manufacturing and automation software engineering. | 2023                       |
| <b>Meritorious Service Medal</b> , ROK Personnel Command<br>Recognition in the K-Startup Contest.                       | May 2020                   |
| <b>Meritorious Service Medal</b> , ROK Cyber Operations Command<br>Recognition for software development contributions.  | Dec 2019                   |
| <b>Dean's List</b> , SUNY Fredonia                                                                                      | May/Dec 2014; May/Dec 2015 |
| <b>Alpha Lambda Delta</b> , National Honor Society for First-Year Students · May 2014                                   |                            |
| <b>SUNY Global Scholarship</b> , SUNY Fredonia · Dec 2013                                                               |                            |

SELECTED CERTIFICATIONS AND TRAINING

RCR Basic Course - CITI Program · Aug 2025  
AWS Partner: Sales Accreditation - Amazon Web Services · Jun 2025  
Self-Driving Cars - University of Toronto · Oct 2023  
Google Project Management - Google · Jun 2023  
Data Intensive Computing - University at Buffalo · May 2019  
Probabilistic Graphical Models - Stanford University · Oct 2019

TECHNICAL SKILLS AND LANGUAGES

**Implementation:** Python, C++, PyTorch, TensorFlow, ROS/ROS2, Isaac Sim, CARLA, React, Django, SQL.  
**Computing:** MPI, NCCL, CUDA, distributed GPU training, Kubernetes, Hadoop, Kafka, Spark.  
**AI systems:** World models, VLA, RAG, MCP, knowledge graphs, Neo4j, LangChain.  
**Languages:** Korean (native/bilingual); English (professional working proficiency).